
Calculating and using sequence-space Jacobians

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Calculating sequence-space Jacobians

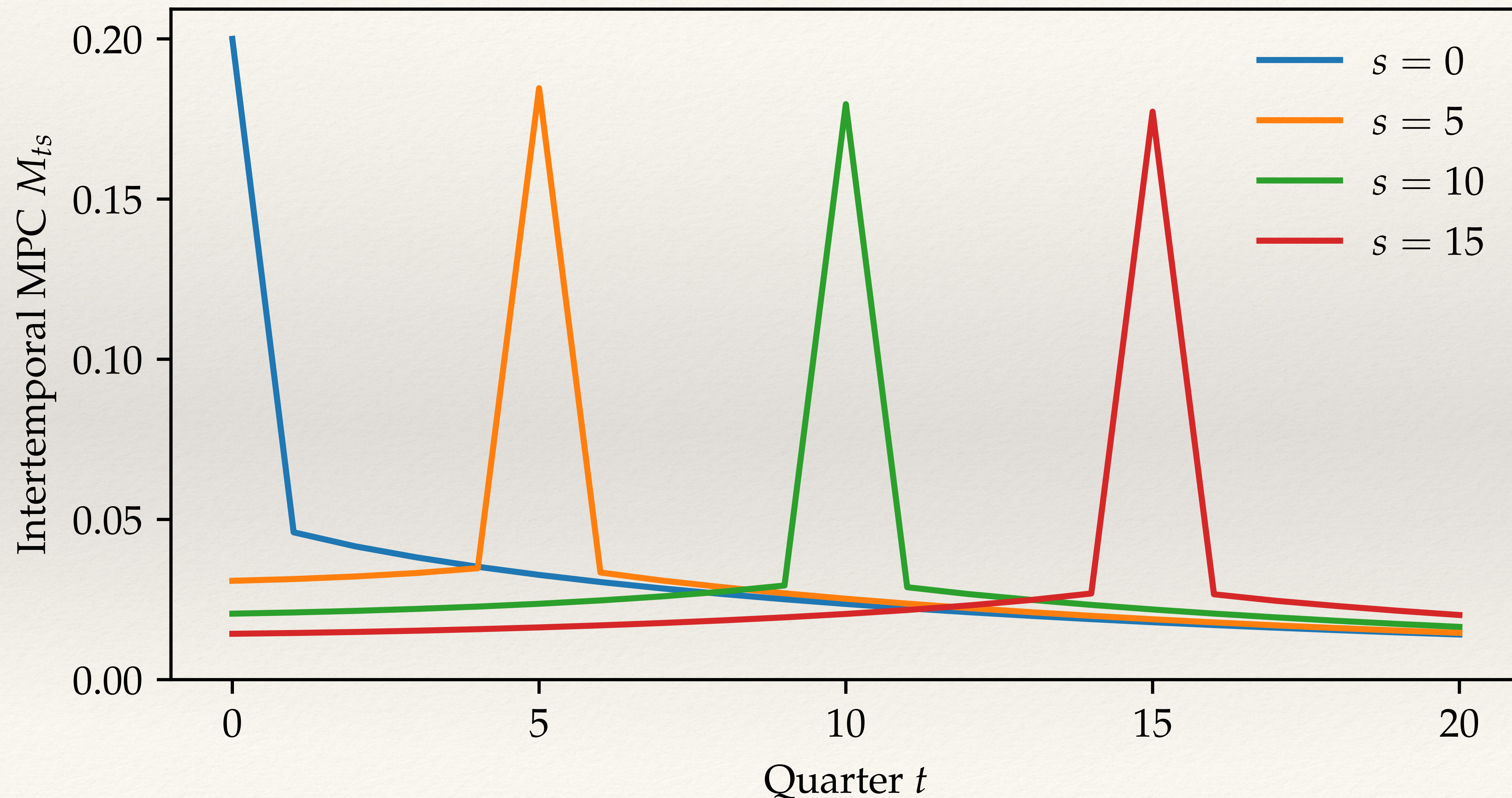
What are sequence-space Jacobians?

- ❖ Take some function mapping sequences of aggregate input to aggregate output
- ❖ Examples so far: consumption and assets as functions of after-tax income Z_t :

$$C_t = \mathcal{C}_t(\{Z_t\}_{t=0}^{\infty}) \quad \text{and} \quad A_t = \mathcal{A}_t(\{Z_t\}_{t=0}^{\infty})$$

- ❖ **Sequence-space Jacobians** are the derivatives of these functions:
 - ❖ Usually around the steady state
 - ❖ In principle, dimensions $\infty \times \infty$, though usually truncate to finite $T \times T$
 - ❖ Can have many, mapping multiple inputs to multiple outputs
- ❖ Example: Jacobian of $\mathcal{C}$ is matrix $\mathbf{M}$ of “intertemporal MPCs”
- ❖ As we saw in fiscal lecture, can use to easily solve for impulse responses

One sequence-space Jacobian: intertemporal MPCs

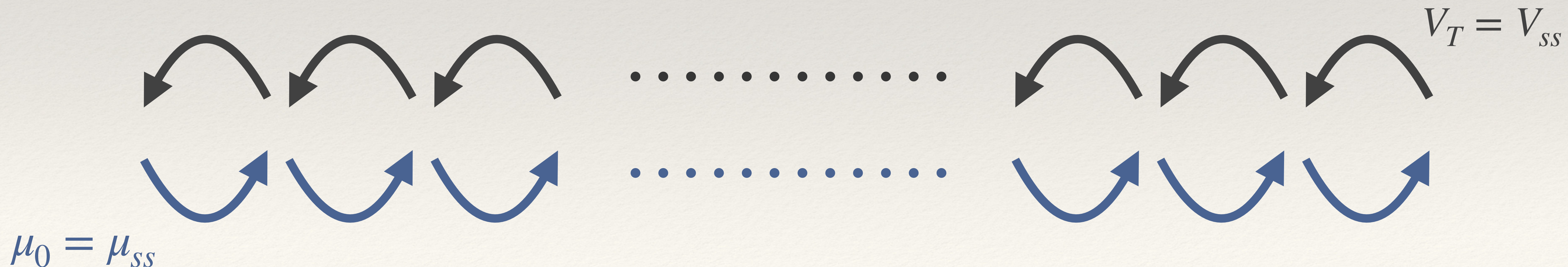


Here we're plotting a few **columns** of the sequence-space Jacobian $\mathbf{M}$, of consumption $\mathcal{C}$

(same $A=100\%$ of GDP calibration as last lecture)

How do we calculate sequence-space Jacobians?

- ❖ Sequence-space Jacobians are awesome if we have them
- ❖ But how do we get them in the first place for a heterogeneous-agent model?
- ❖ Each column is an impulse response to perturbation only at s ...
- ❖ Do we need to redo this process T times, once for each s , at $O(NT^2)$ cost?



[Where “ N ” is dimensionality of het-agent model.]

We can do better

- ❖ The “direct” or “**brute-force**” method is costly:
 - ❖ if $N \gg T$, then $O(NT^2)$ work to get Jacobians swamps $O(T^3)$ cost of matrix operations
 - ❖ (still not totally useless, especially if we can reuse them)
- ❖ Fortunately, there’s a better way: the “**fake news algorithm**”
 - ❖ Need (roughly) *single* backward and forward pass, not one for each s
 - ❖ Reduces bottleneck steps to $O(NT)$
 - ❖ Reference: SSJ paper (Auclert, Bardóczy, Rognlie, Straub, Econometrica 2021)

General setup (similar to SSJ paper notation)

- ❖ Let superscript s denote infinitesimal shock $dZ_s = dx$ at date s
 - ❖ Income at all other dates remains in steady state
- ❖ Iterate backward to get policy functions $\mathbf{c}_t^s$ and transition matrix over discretized states Λ_t^s at each date, represented respectively as a length- N vector and an $N \times N$ matrix
- ❖ Distribution (over discretized states) and aggregate consumption given by

$$\mathbf{D}_{t+1}^s = (\Lambda_t^s)' \mathbf{D}_t^s$$

$$C_t^s = (\mathbf{c}_t^s)' \mathbf{D}_t^s$$

- ❖ Then calculate intertemporal MPCs $M_{t,s} = \partial C_t / \partial Z_s$ as dC_t^s / dx

Insight: only need to iterate backward once!

- ❖ Iterate backward separately to recalculate $\mathbf{c}_t^s$ and Λ_t^s for each s ? **No!**
- ❖ Why? Because **only distance to the shock matters** for policy function:

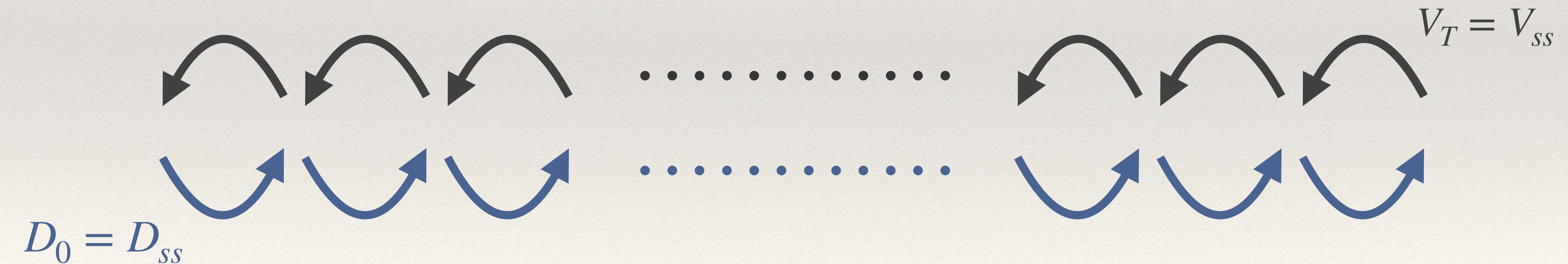
$$\mathbf{c}_t^s = \mathbf{c}_{t+h}^{s+h} \text{ for any } h$$

- ❖ So, just consider one shock at maximal horizon $s = T - 1$, then write (same for Λ)

$$\mathbf{c}_t^s = \begin{cases} \mathbf{c}_{ss} & s < t \\ \mathbf{c}_{T-1-(s-t)}^{T-1} & s \geq t \end{cases}$$

Very helpful, but still lots of work

- ❖ Backward iteration often costliest, so this is a big help!
- ❖ But still, for each s , need to iterate forward on distribution
- ❖ Economized on top steps but not bottom:



What's going on?

- ❖ We care about aggregate $C_t^s = (\mathbf{c}_t^s)' \mathbf{D}_t^s$ [or, more specifically, $M_{t,s} \equiv dC_t^s/dx$]
- ❖ We have $\mathbf{c}_t^s = \mathbf{c}_{t+h}^{s+h}$, but that's not true for $\mathbf{D}_t^s$: generally $\mathbf{D}_t^s \neq \mathbf{D}_{t+h}^{s+h}$
- ❖ Theorem: to first order,

$$d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1} = (\Lambda'_{ss})^{t-1} d\mathbf{D}_1^s$$

- ❖ Why? If shock happens at s instead of $s - 1$, **one more period to anticipate it**
 - ❖ $\rightarrow$ affects date 0 policy $\rightarrow$ affects distribution date-1 distribution $d\mathbf{D}_1^s$
 - ❖ $\rightarrow$ carries over to date t distribution via $t - 1$ applications of Λ'_{ss}

Effect on aggregates

- ❖ We have $d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1} = (\Lambda'_{ss})^{t-1} d\mathbf{D}_1^s$
- ❖ Effect on $dC_t^s - dC_{t-1}^{s-1} = \mathbf{c}'_{ss}(d\mathbf{D}_t^s - d\mathbf{D}_{t-1}^{s-1})$ is therefore:

$$\mathbf{c}'_{ss}(\Lambda'_{ss})^{t-1} d\mathbf{D}_1^s \quad (\equiv F_{t,s} \cdot dx)$$

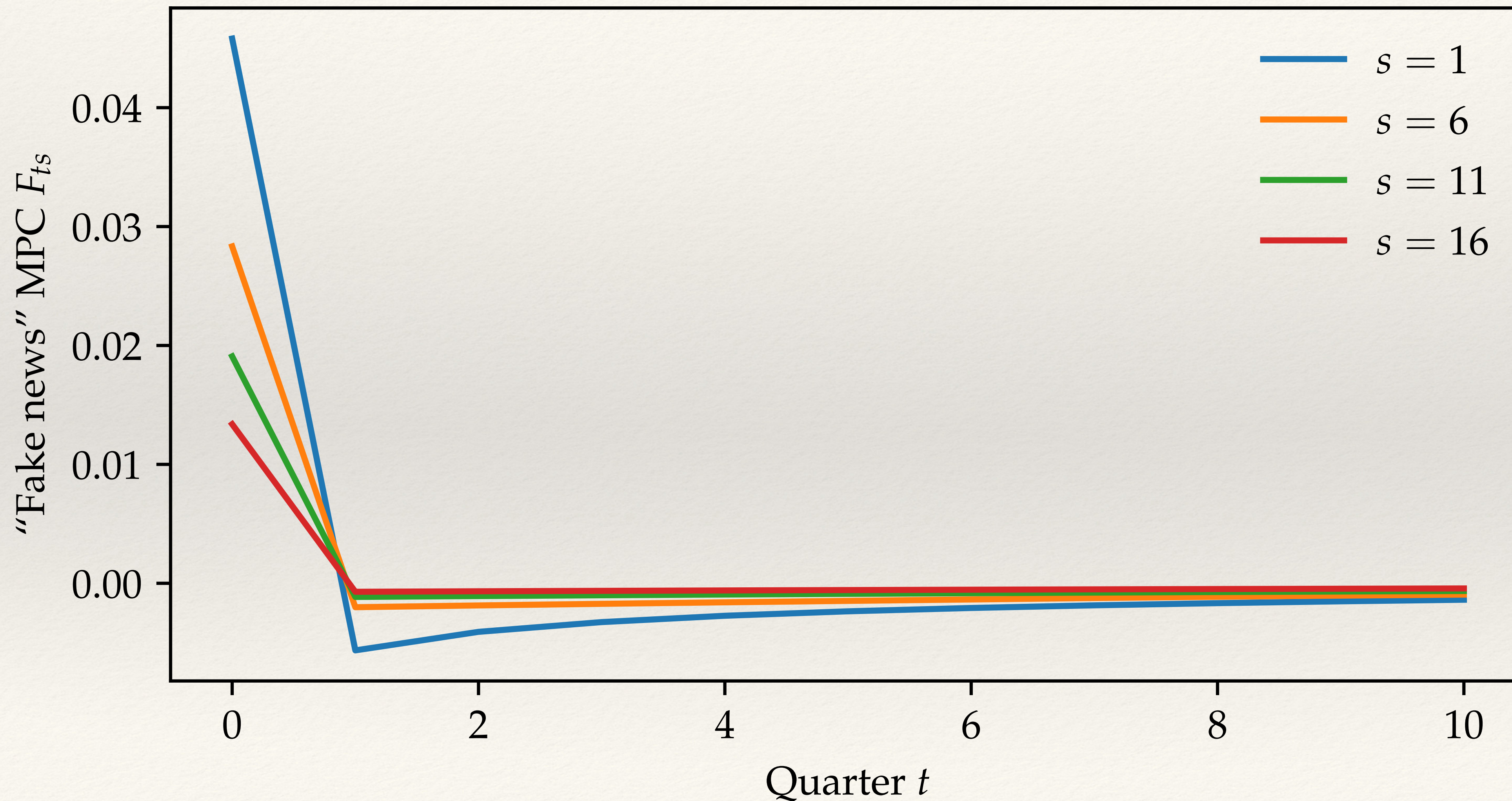
- ❖ The matrix $F_{t,s}$ is closely related to Jacobian $M_{t,s}$ via $F_{t,s} = M_{t,s} - M_{t-1,s-1}$
- ❖ Can reconstruct $M_{t,s}$ from diagonals $F_{t,s}$ (defining $F_{t,s} \equiv M_{t,s}$ for t or $s = 0$):

$$M_{3,4} = F_{3,4} + F_{2,3} + F_{1,2} + F_{0,1}$$

What is this F (“fake news matrix”)?

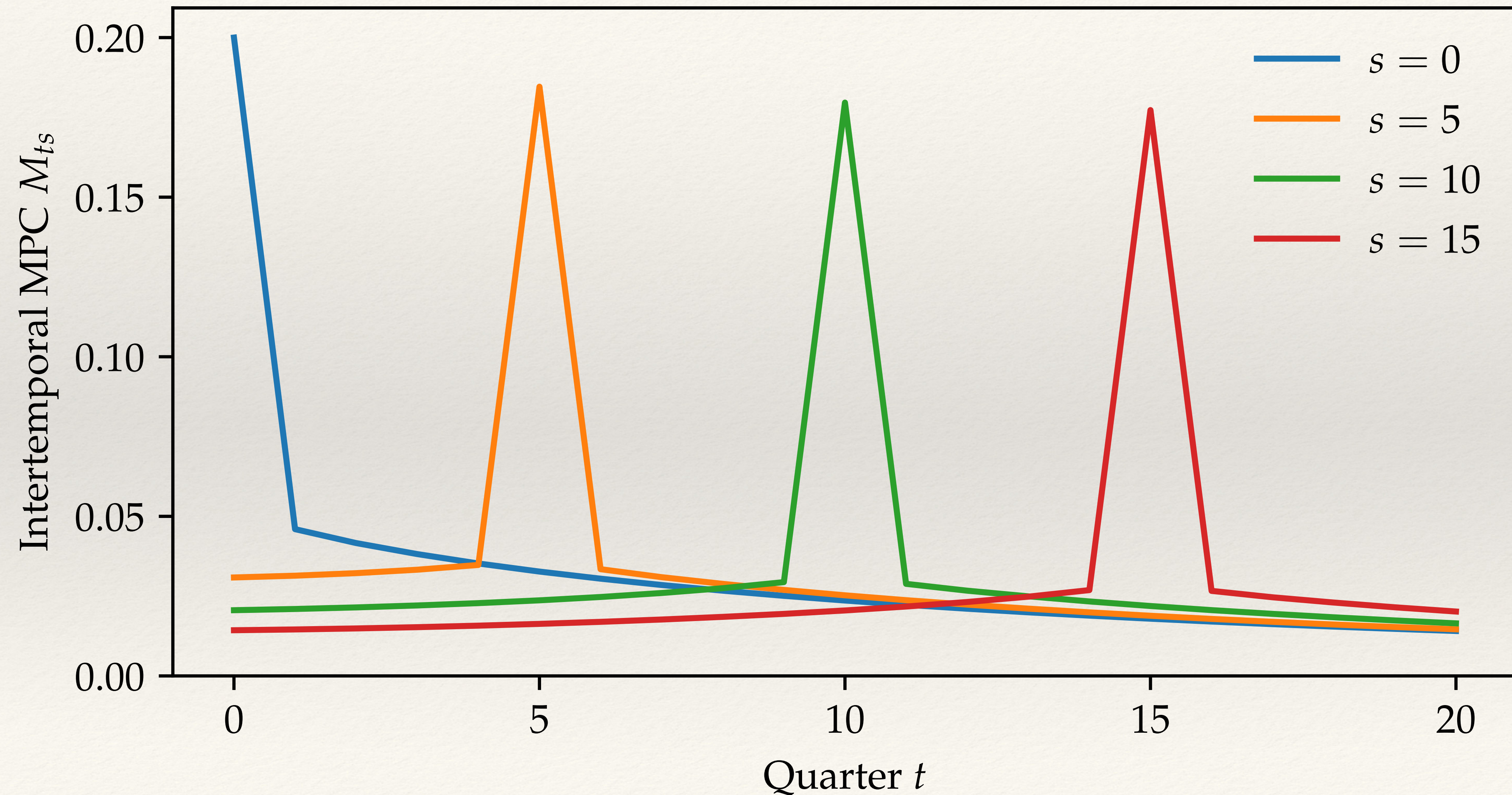
- ❖ For $t, s > 0$, we have $F_{t,s} = M_{t,s} - M_{t-1,s-1}$
- ❖ Why are $M_{t,s}$ and $M_{t-1,s-1}$ different?
 - ❖ Because former has **one extra period of anticipation**
 - ❖ $F_{t,s}$ is the effect at t of having thought, at 0, that there would be shock at s
- ❖ One interpretation: “fake news shock”
 - ❖ $F_{\cdot,s}$ is impulse response to shock at s announced at 0, rescinded at 1

Visualizing columns of F

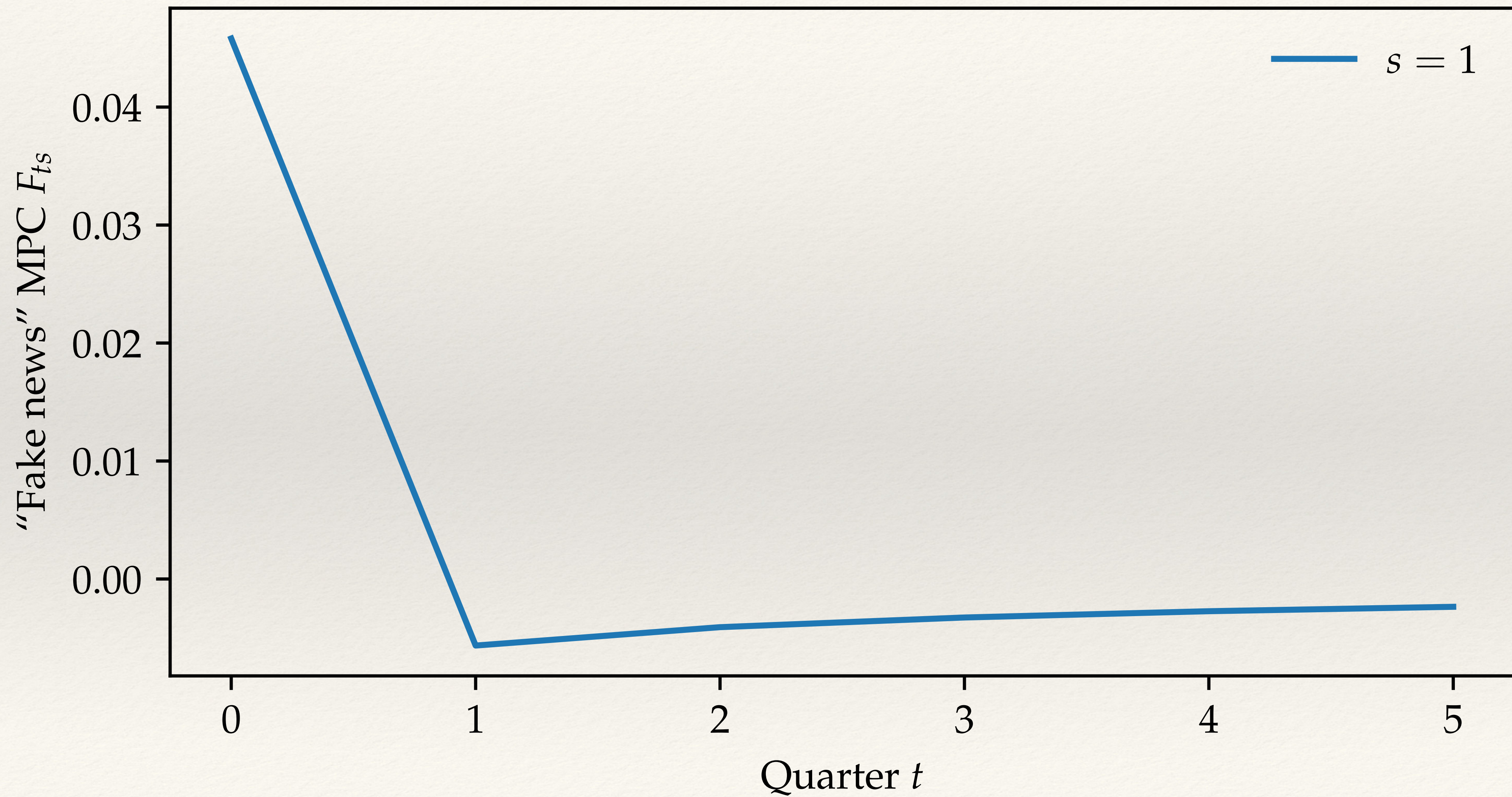


In response to
"fake news
shock" about
future income,
you consume
more at date 0,
then have
fewer
resources
going forward

Difference between $M_{t,1}$ and $M_{t-1,0}\dots$



... is exactly $F_{t,1}$



Where we stand now

- ❖ Reduced finding Jacobian $\mathbf{M}$ to “fake news matrix” $\mathbf{F}$
- ❖ Simple formula for $F_{t,s}$ when $t > 0$:

$$F_{t,s}dx = \mathbf{c}'_{ss}(\Lambda'_{ss})^{t-1}d\mathbf{D}_1^s$$

- ❖ Problem: still seems like a lot of work to apply Λ'_{ss} repeatedly to each $d\mathbf{D}_1^s$!
- ❖ Solution: evaluate formula from the **left, not the right!**
- ❖ Calculate “**expectation functions**” $\mathcal{E}_t \equiv (\Lambda_{ss})^t \mathbf{c}_{ss}$ **only once**, then evaluate $\mathcal{E}'_t d\mathbf{D}_1^s$
 - ❖ $\mathcal{E}_t$ is expected c in t periods for a household who follows steady-state policy

Cracked it open, now have four-step algorithm

- ❖ **Step 1:** iterate backward once from shock $dZ_{T-1} = dx$ to obtain all $\Lambda_t^s, \mathbf{y}_t^s$
 - ❖ define $\mathcal{Y}_s dx \equiv (d\mathbf{c}_0^s)' \mathbf{D}_{ss}$ and $\mathcal{D}_s dx \equiv (d\Lambda_0^s)' \mathbf{D}_{ss}$
- ❖ **Step 2:** repeatedly apply Λ_{ss} to calculate expectation functions $\mathcal{E}_t \equiv (\Lambda_{ss})^t \mathbf{c}_{ss}$
- ❖ **Step 3:** form fake news matrix, which is $F_{0,s} = \mathcal{Y}_s$ and $F_{t,s} = \mathcal{E}_{t-1}' \mathcal{D}_s$ ($t > 0$)
- ❖ **Step 4:** calculate all $M_{t,s}$ by cumulatively summing diagonals of $F_{t,s}$
- ❖ First 2 steps are $O(NT)$, step 3 is $O(NT^2)$ but can be written as giant matrix multiplication (super efficient, never the bottleneck), step 4 is $O(T^2)$

Summary: the “fake news algorithm”

- ❖ Most complex of ideas so far, but now sequence-space Jacobians are practical!
 - ❖ Key step is only $O(NT)$, far better than the $O(N^3)$ of state-space methods
 - ❖ Example was iMPCs $\mathbf{M}$, but same method for any other Jacobian
 - ❖ Various implementation details (for multiple inputs / outputs, numerical vs. automatic differentiation, ...): see SSJ paper and appendix
- ❖ Reducing Jacobians to “fake news matrices” an interesting step in own right
 - ❖ Isolate effects of information, useful for deviations from FIRE

Using sequence-space Jacobians

Linear and nonlinear systems

- ❖ Asset version of the Intertemporal Keynesian Cross model (last lecture):

$$\mathcal{A}_t \left(\{Y_s - T_s\}_{s=0}^{\infty} \right) = B_t$$

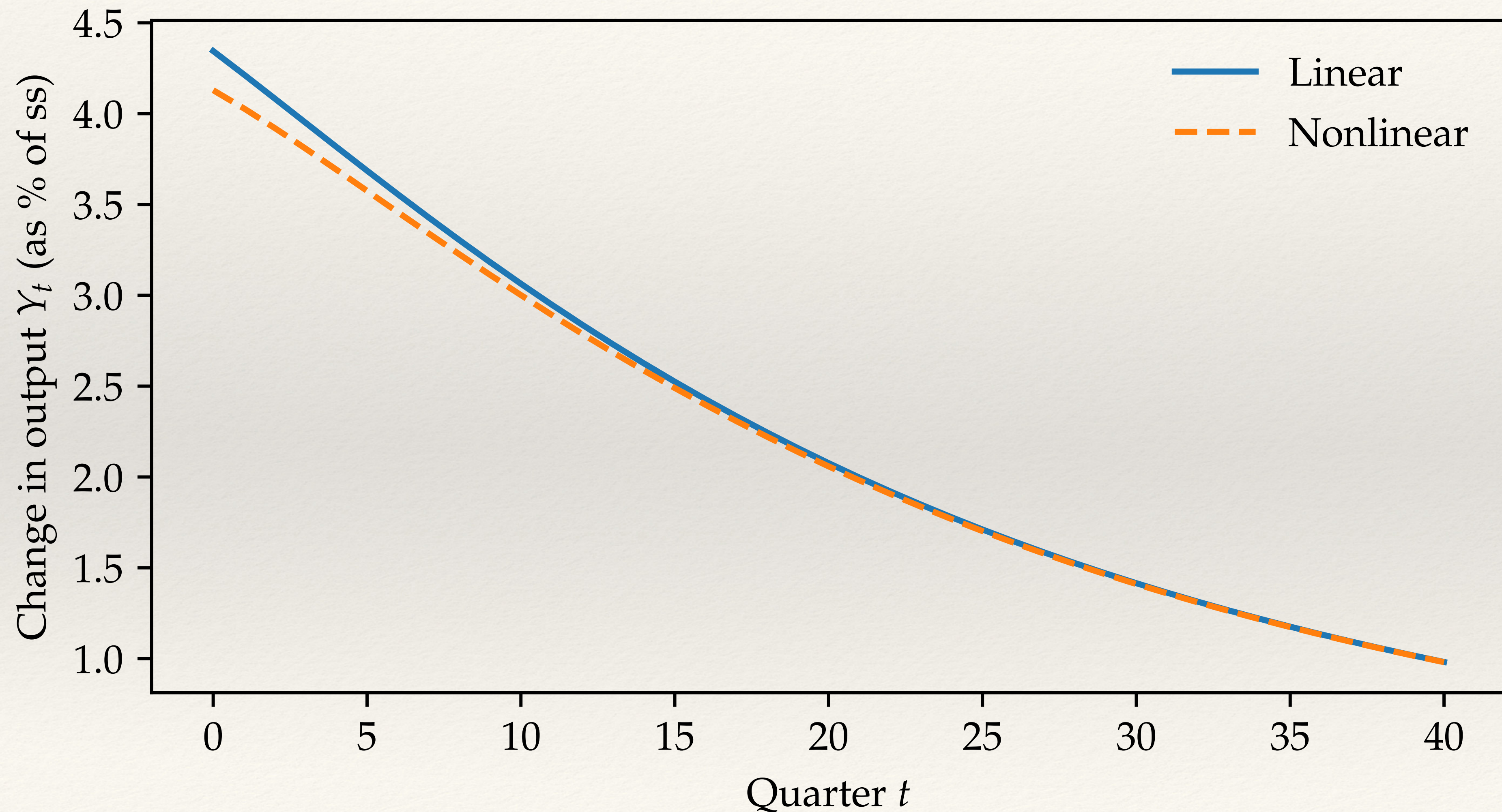
- ❖ Solve **linearized** case with simple matrix operations, where $\mathbf{A}$ is SSJ of $\mathcal{A}$:

$$\mathbf{A} \left(d\mathbf{Y} - d\mathbf{T} \right) = d\mathbf{B} \quad \Rightarrow \quad d\mathbf{Y} = d\mathbf{T} + \mathbf{A}^{-1} d\mathbf{B}$$

- ❖ For a **nonlinear** impulse response, solve using version of Newton's method, using $\mathbf{A}$ as approximate Jacobian even around nonlinear path
- ❖ Iterate over guesses $\mathbf{Y}^{(j)}$ until asset market clearing approximately holds:

$$\mathbf{Y}^{(j)} = \mathbf{Y}^{(j-1)} - \mathbf{A}^{-1} \cdot (\mathcal{A}(\mathbf{Y}^{(j-1)}) - \mathbf{T}) - \mathbf{B}$$

Result when scaling tax cut to 5% of GDP



Use same tax cut experiment as in last lecture, but scale impact tax cut to 5% of GDP and compare linear vs. nonlinear (huge tax cut!)

Slightly smaller nonlinear response, due to concavity of consumption function

Advantages of the sequence space and Jacobians

Any shock, any heterogeneity

- ❖ For linear case, we solved

$$d\mathbf{Y} = \mathbf{A}^{-1}d\mathbf{B} + d\mathbf{T}$$

- ❖ Once we've calculated inverse asset Jacobian $\mathbf{A}^{-1}$, we can solve for response to **any time path** of $d\mathbf{B}$ and $d\mathbf{T}$ almost instantly
- ❖ Similar “general equilibrium Jacobian” mapping in more complex cases
- ❖ Suppose we want 100 different types, over and above our heterogeneity
 - ❖ Just take weighted average of the $\mathbf{A}$ s to get economy-wide $\mathbf{A}$

Some advantages of the sequence space: summary

1. Can get response to **any shock**
2. Can easily handle almost **any heterogeneity**
3. Can simulate, get **any second** moments, even **estimate model** [to come!]
4. Can implement **non-rational expectations** [to come!]
5. Can get **informative decompositions** [e.g. $d\mathbf{Y} = d\mathbf{G} - \mathbf{M}d\mathbf{T} + \mathbf{M}d\mathbf{Y}$]

(These advantages carry over in part to any MIT shock / sequence-space method, but best by far when we're using Jacobians!)